OpenIllusionist *(https://www.openillusionist.org.uk/home/):*

- Dynamic integration of objects
- Real-time user inputs
- Dynamic modelling
- Virtual objects analytics

ApertusVR *(http://apertusvr.org/):*

- High performance platform for AR, VR and mixed reality (MR)
- HTTP rest API integrated
- Embeddable framework-independent
- Network topology-independent

ARToolkit+ *(http://www.arreverie.com/artoolkit.html):*

- Modular open source code
- Smart glasses
- Content library
- Mobile focus on secure development
- Android and iOS support
- Content creation
- Simulation
- Superimposed objects
- Drag-and-drop
- Virtual reality integration

Holokit *(https://holokit.io/):*

- Virtual reality integration
- Real-world backdrop
- Superimposed objects
- Inside-out tracking function
- 3D objects
- Multi-degree diagonal field of view
- Content creation
- Merges the real and virtual

BRIO *(https://experience.brioxr.com/):*

- Drag-and-drop interface
- Multiple file format support
- Virtual reality integration
- 3D objects
- Content creation
- Cross-device publishing
- Content library
- Real-world backdrop
- Simulation
- Tracking and analytics
- Superimposed objects

Figure 2: Augmented and virtual reality

Adobe Aero *(https://www.adobe.com/in/products/aero.html):*

- Dynamic interactive elements
- Content creation
- Set pathways for animations
- Drag-and-drop
- Content library
- Tracking and analytics
- Simulation
- Virtual reality integration
- Cross-device publishing

Vuforia Engine *(https://developer.vuforia.com/):*

- Unparalleled accuracy
- Dynamic simulation
- Virtual reality integration
- 3D objects
- Creative empowerment
- Superimposed objects
- Content creation
- Maximum reach
- Content library
- Tracking and analytics
- Real-world backdrop
- Advanced vision
- Drag-and-drop

Scope for research and development

AR and VR help in providing a multisensory experience in which real-world items are improved with computer-generated perceptual information, including visual, aural, haptic, somatosensory, and olfactory information. Combining the real and virtual worlds, these technologies allow in-the-moment interaction, and accurately register 3D models of virtual and actual items.

Researchers and practitioners can use open source platforms of AR and VR to provide an interface between real-time experiences and remote infrastructure. There is a huge scope for research as there is no need to move physically to any location and remote points are always accessible.

Scientists, medical practitioners, surgeons, and experts in space exploration missions can use AR, VR and MR (mixed reality) to have a simulated view or remote experience of the actual infrastructure. This can be beneficial for multiple applications. END

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